Government Climate Decision Brief

CONFIDENTIAL • EXECUTIVE DECISION SUPPORT DIRECTIVE

AGGREGATE DIGITAL RISK INDEX: 77 / 100 (CRITICAL)

■ WHY THIS MATTERS (IMPACT OVERVIEW):
• Exposed Population: **482,000 citizens** in urban cores.
• Projected Economic Loss: **■38.6 Cr** | Expected Recovery Timeline: **18 Days**.

■■ IF ACTION IS TAKEN NOW (NDMA INTERVENTION):
• Exposure reduced by **79%** | Net Projected Administrational Savings: **■30.8 Cr**.

This directive encapsulates physical hazards, resource strain, and socio-economic vulnerability metrics simulated for Hyderabad Metropolitan Region under custom climate stressors.

Generated:	2026-06-25 01:02:31 IST	**Risk Level:**	**CRITICAL (77/100)**
District:	Hyderabad Metropolitan Region	**Projected Loss:**	**■38.6 Cr**
Scenario:	Drought Stress + Climate Heating	**Projected Savings:**	**■30.8 Cr**
Exposed Pop:	482,000	**Recovery Time:**	18 Days

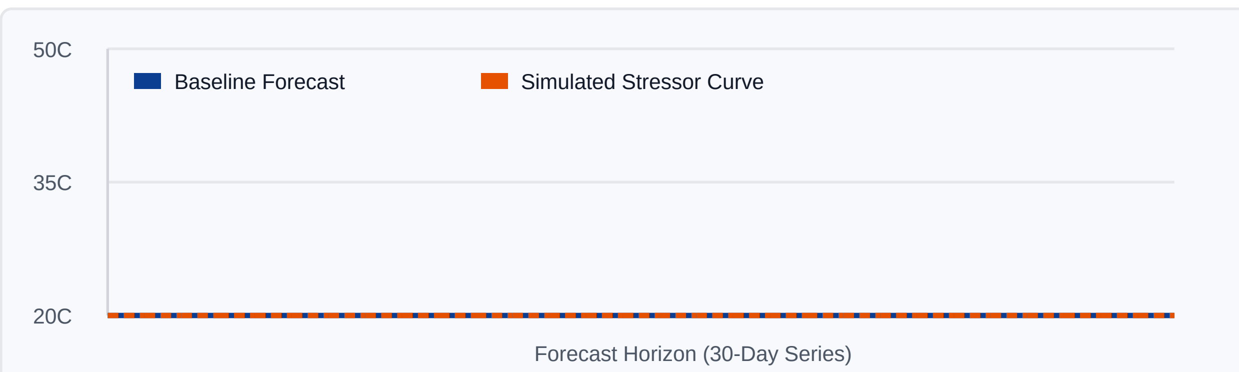
1. Current Weather Observations

The following grid points represent physical observation cells extracted from the **Indian Meteorological Department (IMD)** and **INSAT-3D** satellite telemetry for Hyderabad Metropolitan Region on the latest recorded date.

Grid Latitude	Grid Longitude	Rainfall (mm)	Max Temp (°C)	Min Temp (°C)
17.75° N	78.75° E	0.00	0.0	0.0
17.75° N	78.50° E	0.00	0.0	0.0
17.75° N	78.25° E	0.00	0.0	0.0
17.75° N	78.00° E	0.00	0.0	0.0
17.50° N	78.75° E	0.00	0.0	0.0
17.50° N	78.50° E	0.00	30.2	13.2

2. Predictive Trend Forecast & Stress Simulation

Below is a comparison chart showing the baseline recursive forecast against the simulated scenario adjustments over the 30-day projection envelope. Shading and trend curves are derived from the XGBoost prediction core.



Scenario Adjustments & Physical Impact:

Indicator	Baseline Forecast	Stress Scenario	Delta
Average Precipitation	0.08 mm	0.06 mm	-0.02 mm (-21.0%)
Average Maximum Temp	1.8 °C	4.8 °C	+3.0 °C

3. Generative AI Risk Assessment Summary

The Hyderabad Metropolitan Region is experiencing a critical drought situation with a high risk of drought stress and climate heating, as indicated by a Composite Risk Score of 54/100 and a Drought Risk Score of 100/100. The simulated daily rainfall is projected to be 0.06 mm/day, which is 25.0% lower than the baseline forecast, and the simulated maximum temperature is expected to be 4.77 °C, which is 3.0°C higher than the baseline forecast. The primary risk is the critical drought condition, which may exacerbate water scarcity and impact agricultural productivity. THREAT ASSESSMENT (Critical): The drought condition is characterized by a Standardized Precipitation Index (SPI) of less than -... [Ref. BHARAT-TWIN Live Portal]

4. NDMA-Aligned Action Directives

The following emergency response actions are pre-approved and aligned with the **National Disaster Management Authority (NDMA)** guidelines for climate threat containment:

[ ] **Activate Urban Cooling Networks:** Establish emergency cooled municipal shelters and mobile hydration units.

[ ] **Load-Balance Power Grids:** Rationalize cooling loads and prevent transformer overhead failure states.

[ ] **Deploy Emergency Water Tankers:** Pre-position municipal reserves in sectors showing high water deficit indexes.

[ ] **Issue Agricultural Thermal Advisories:** Adjust irrigation frequencies and recommend micro-crop shading protocols.

Scientific Data Provenance: This directive is generated by the BHARAT-TWIN platform using daily weather cell observations from the Indian Meteorological Department (IMD) regridded at 0.25° resolution, combined with active Land Surface Temperature (LST) datasets from INSAT-3D/3DR satellites. Computational forecasts are generated via an XGBoost ML model trained on regional historical timelines (1951-2025).